

Moxon-Yagi Antenna

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This report analyzes the performance of a dual-band Moxon-Yagi antenna, which combines two independent Moxon and Yagi designs in order to achieve multiple-frequency resonance.

1. INTRODUCTION

The Moxon-Yagi antenna is a dual-band VHF/UHF resonator with high directivity, making it an ideal candidate for satellite communication of uplink/downlink frequencies over large distances. The benefit of the Moxon-Yagi design over other directional antennas is its electrical simplicity. This antenna requires only 1 feed line and no diplexers while operating both 2 meter and 70 centimeter bands efficiently. It radiates at "end-fire", meaning that the electromagnetic wave produced propagates along the same axis as its dipole components. The design is accredited to call-sign LY3LP, and this report is for the purpose of recreating and analyzing its performance.

2. HOW IT WORKS

The Moxon components (5 & 7) make up the 2m / VHF resonance system. The design was first introduced by Leslie Moxon as the "Two-Element Driven array". Element 5 is referred to as "driven" because of its direct connection with the TX/RX line. Element 7 is the "reflector" component, creating a conductive bound where all EM waves produced at the back end of the driven element can be reflected in the forward direction.

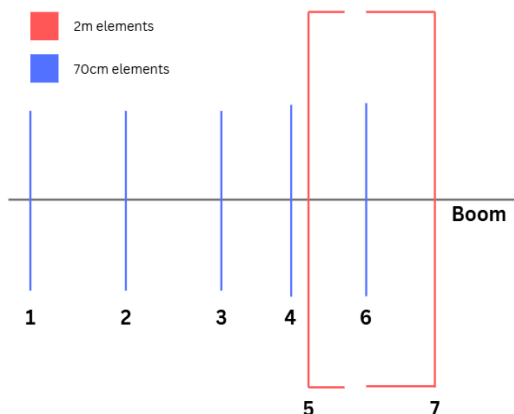


FIG. 1: Geometric layout of antenna design with number designations to each element for future reference. Elements 1,2,3,4, and 6 comprise the 70cm resonating system, while 5 and 7 create the 2m resonating system. Feed-line is located at the cross section of the boom and element 5.

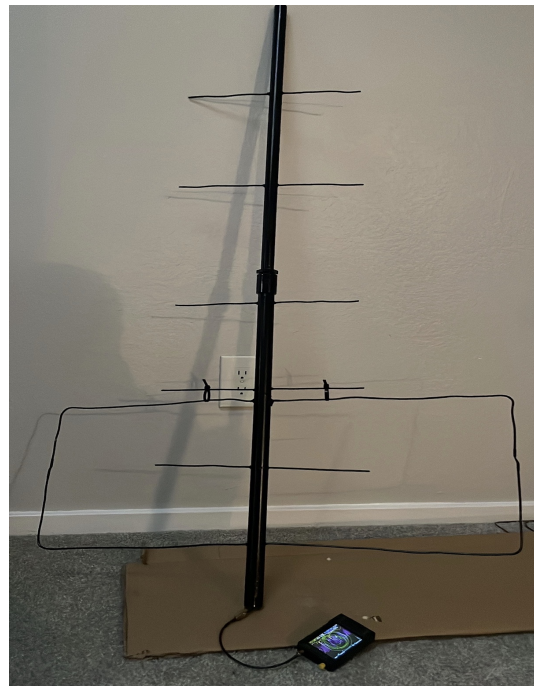


FIG. 2: Fully assembled Moxon-Yagi, used for proof of concept and performance measurements

What sets it apart from the typical Yagi antenna is the bent edges of each dipole, which introduces a mechanically controlled capacitive load and reduces the need for additional matching networks. Capacitive loads bring down resonance frequency, allowing the dipole to be physically shorter than half wavelength. The geometry also increases the forward-to-back ratio by encapsulating the fields produced by the driver rather than letting them freely radiate at the edges.

The reflector is generally larger than the driven element (5 %) to "catch" any stray power effectively. The exact length isn't as vital as the driven element for an exact resonance frequency. The 70 cm Yagi components aren't directly connected to any driven elements, but instead utilize passive coupling.

The primary passively coupled element (4) ranges from $.45 \lambda$ to $.48 \lambda$ to compensate for parasitic effects. The reflector (6) should again be at least 5% larger. The directive components (1, 2, 3, and 4) should be separated by 0.15λ to 0.3λ . The addition of the first driven element predictively results in an improvement of approximately 7dB, and each additional element about 1dB.

3. MATERIALS

- 8-Gauge Solid copper wire (12 ft or more)
- PVC pipe 1 in X 24 in
- PVC pipe 3/4 in x 24 in
- 1 in to 3/4 in PVC coupling
- Gorilla Glue
- 4 Heat Shrinks

Expected Cost: 40 \$

Optional:

- Zip Ties
- Enamel Gloss Spray Paint

Equipment:

- Nano-VNA
- Power Drill
- Soldering Iron
- Wire Cutters

4. ASSEMBLY

It is required to have at minimum 12 ft of copper wire; however, I would suggest 15 ft to give ample room for error and reconstructing. The PVC pipe used for the boom should be at least 1 meter in length and its diameter is largely negligible. There are trade-offs between having a smaller diameter that offers less interference or a larger diameter that provides more stability in the balancing of copper wire.

Step 1: Preparation cut wires into appropriate length segments and straighten. *Tip: cut slightly above the desired length and then trim for precise results. If the wire is difficult to straighten manually, connect one end to a power drill and pull it at the other end while it spins. This creates tension in the wire. It is easiest to do once the smaller segments are cut.*

Step 2: Connections strip the coax of its outer and inner layer, cut element 5 evenly in half and solder each. Heat shrinks are added to cover each bare connection. This prevents unwanted interference. Now you can bend the edges of each corner and test your first element with a Nano-VNA to ensure resonance. If its higher than your expected frequency, trim edges and change bending points

Step 3: Drill holes in PVC corresponding to appropriate lengths between wires. For the feed component, you should drill a slightly larger hole on one side to insert coax cable. *Tip: use a string around the pipe and mark*

at half length to ensure holes drilled create a path directly through pipe.

Step 4: insert wires through holes in the pipe and measure each side to ensure symmetry. Add glue to holes so that the wires are secure. Fill the gap where coaxial cable was inserted with any material low in interference. I would recommend a rolled piece of printer paper. *Tip: use a book to hold up sides while drying to make sure all elements dry without falling out of line with one another. Ensuring all elements are in the same plane ensures efficiency in the antenna. (especially on larger elements that are more prone to rotations and tilt)*

Step 5: add heat shrinks to gaps between element 5 and 7 for support and stability in gap length.

5. RESULTS

Local return loss minimums occurred at 142 MHz and 427 MHz, with impedances of 56.27 - j0.41 ohms and 50.03 + j2.07 ohms respectively [fig : 3]. S11 measurements agreed with resonance points, delivering a reflection loss of 12.9 dB at VHF and 24.6 dB at UHF. These dips in return loss seen indicates a higher power transfer between the antenna and free space (resonance). This assumes minimal contributions to thermal loss or internal dissipation, which traditionally don't show such a strong frequency dependence. Bandwidth appears to be 10 MHz and VHF and 5 MHz at UHF. This agrees with theoretical assumptions of the singular Moxon (2m) antenna having a larger bandwidth than a Yagi. One can attribute this to the notion of mutual coupling. The folded dipole used in the Moxon contains two parallel wires at the edges that introduce capacitive loads. As frequency increases, inductive impedance occurs which is counteracted by the capacitive, keeping the antenna impedance closer to 50 ohms over a longer band. From bandwidth observations, we can determine that the quality factor of a Yagi is two times higher than that of a Moxon antenna [eq : 1].

$$Q = f_r / \beta \quad (1)$$

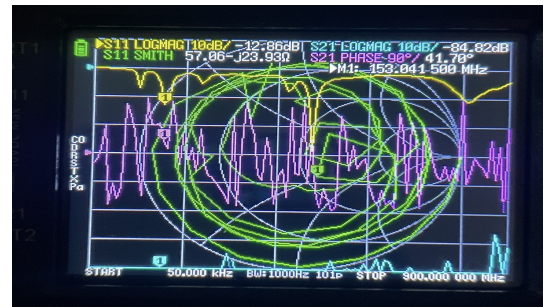


FIG. 3: Measured return loss of antenna over frequency sweep (12.9 dB @ 142 MHz and 24.6 dB @ 427 MHz).

6. POINTS OF ERROR

The VNA calibration plane began before the coaxial cable that fed the active element, as opposed to the end. Approximately 1.5 dB of loss should be removed from the total Return Loss measured. Traditionally, a precise measurements would require the characterization of the cable itself, to mathematically remove its contributions to overall s-parameters. However, the stripping and soldering of the coaxial cable in the direct connection to the driven element prevented this step. An alternative feed design could be proposed to improve calibration. The structure itself was imperfect in geometry. Thick copper wire, while moldable, wasn't perfectly straightened throughout individual components. While I did my best to keep all elements perfectly aligned in an xy plane, there are undoubtedly slight offsets in rotation which provide space for RF power leakages. All of these variations can contribute to less efficient performance with respect to simulations.

7. CONCLUSION

The performance of this antenna is adequate for 2 m/70 cm band radio communication. While there are much easier assemblies for dual band Yagis available online at roughly 100 dollars, I would encourage readers to self-assemble. It is cost advantageous, especially if you wish to pursue a hobby in HAM radio where you can reuse required equipment and material. Self-assembly also leaves room for freedom in parameters. The mechanical tuning component and high spatial variance allows the hobbyist to make their own adjustments and see first-hand how the antenna performs as a result.

8. MISC. FIGURES

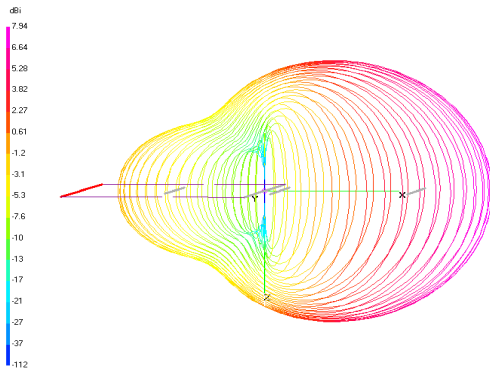


FIG. 4: 3D depiction of Moxon-Yagi Radiation pattern at 140 MHz using AN-SOF software

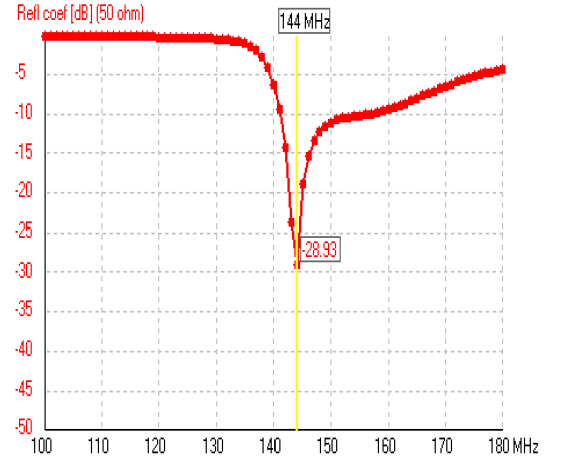


FIG. 5: Simulated return loss of antenna at 144 MHz

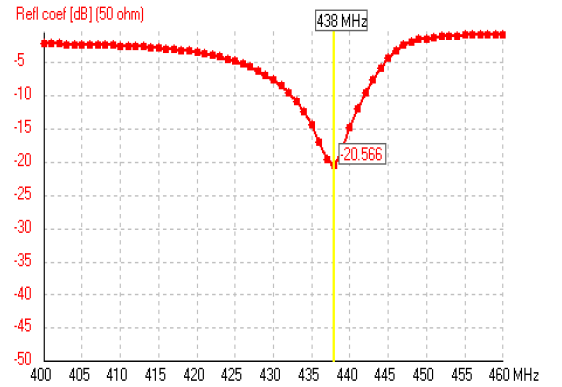


FIG. 6: Simulated return loss of antenna at 438 MHz

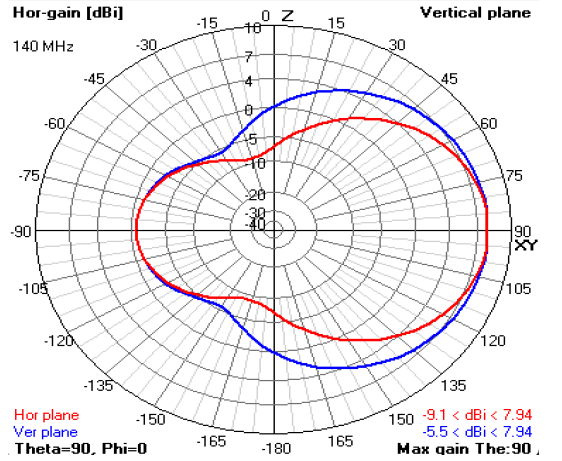


FIG. 7: H and V cuts of radiation patterns at 140 MHz using AN-SOF software. Notably, the vertical plane has a much larger beamwidth than the horizontal plane. This can be attributed to the fact that the antenna has more surface area in the horizontal plane, sharpening it's directivity.

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- [1] Philippine Amateur Radio Association, "Making a Dual-Band Satellite Antenna." Available: <https://www.para.org.ph/para-making-a-dual-band-satellite-antenna.html>.
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- [4] Tony Golden, "The Moxon-Yagi Dual-Band VHF/UHF Antenna for Superior Satellite Link Performance." Available: <https://antennasimulator.com/index.php/knowledge-base/moxon-yagi-dual-band-vhf-uhf-satellite-antenna/>.